Stochastic Volatility Models

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Introduction

- Dupire result essentially states that perfect calibration is feasible within local volatility (LV) models.
- However LV models are able to match the value of the smile **as of today**, but how about tomorrow? Or in a week? or in a month?
- LV can be very poor in capturing the **dynamics** of the smile: that is today you have perfect calibration but the fit to tomorrow'd surface is no good (see Dumas, Fleming and Whaley (1998))
- So one needs to re-fit frequently the LV obtaining highly unstable volatility coefficients.
- Also LV models are poor in capturing the volatility dynamics. It seems rather unrealistic that the volatility's movements are so tightly connected with the movements of the underlying.
- Stochastic volatility models - where the volatility in not uniquely determined by the level of the underlying - seem to offer a more plausible/realistic alternative to the LV models.

SV models: the Valuation equation, JG Chapter 1

The Stochastic Volatility framework

- Also in this case we consider a market with two assets: a risk-less asset with constant interest rate r and a risky asset
- The dynamics of the risky asset takes the following form

$$
\begin{aligned}
dS_t &= \mu_t S_t dt + \sqrt{v_t} S_t dZ_t^1 && (1) \\
dv_t &= \alpha(S_t, v_t, t)dt + \eta\beta(S_t, v_t, t)\sqrt{v_t} dZ_t^2 && (2)
\end{aligned}
$$

  where μ_t is the (deterministic) expected rate of return, and Z^1 and Z^2 are correlated Brownian motions with correlation $dZ_t^1 dZ_t^2 = \rho dt$, with ρ constant.

- We also assume that the underlying asset pays no dividends.
- v represents the instantaneous variance - square of the volatility process - of the log-returns: It is described by a diffusion process with its own source of randomness Z^2 - possibly correlated but not perfectly determined by the movements of the underlying asset S.

- Let's focus for a second on the variance process

$$dv_t = \alpha(S_t, v_t, t)dt + \eta\beta(S_t, v_t, t)\sqrt{v_t}dZ_t^2$$

- The drift coefficient $\alpha(S_t, v_t, t)$ typically displays some sort of mean reversion feature, the most known example being

$$\alpha(S_t, v_t, t) = \lambda(\theta - v_t)$$

where λ and θ are positive constants.

- AS for the diffusion coefficient, the particular parametrization is chosen so that $\eta\beta(S_t, v_t, t)$ represent the standard deviation of changes in volatility- the so-called vol-of-vol.
- JG employs the parametrization $\eta\beta(S_t, v_t, t)$ to make sure that there is a parameter governing the scale of the vol-of-vol - i.e. the parameter η. In particular as $\eta \to 0$ the volatility tends to a deterministic function.
- However this is a bit redundant at this stage and at some point it disappears from his derivations. So we right away incorporate η into the function β and we work with the following formulation

$$dv_t = \alpha(S_t, v_t, t)dt + \beta(S_t, v_t, t)\sqrt{v_t}dZ_t^2 \tag{3}$$

rather than (2).

- Once again our problem is how to price a simple claim - such a European call - under the new market dynamics.
- We shall try to follow the same strategy adopted in the LV market model - and try to form a synthetic bank account/ risk-less portfolio. However there are some complications
- The process S itself is not markovian any longer, but the couple (S, v) is markovian: therefore it is reasonable to assume that the price process of the derivative takes the form $V(S_t, v_t, t)$ where V is some smooth function.
- Secondly, when forming the candidate risk-less portfolio we cannot take a position uniquely in the given derivative and the underlying: this wouldn't be enough to eliminate both sources of risk dZ^1 and dZ^2.
- We need to incorporate an extra asset/derivative - say with value process $V^1(S_t, v_t, t)$ and form the following the following portfolio

$$\Pi = V - \Delta S - \Delta^1 V^1$$

where Δ and Δ^1 need to be determined to make Π riskless.

The result

In the market model

$$
\begin{aligned}
dS_t &= \mu_t S_t dt + \sqrt{v_t} S_t dZ_t^1 && (4)\\
dv_t &= \alpha(S_t, v_t, t)dt + \beta(S_t, v_t, t)\sqrt{v_t} dZ_t^2 && (5)
\end{aligned}
$$

an arbitrage-free price process for a simple claim with payoff function $\varphi(S_T, v_T)$ takes the form $V(S_t, v_t, t)$ where V is the solution of the bvp

$$
\begin{aligned}
\frac{\partial V}{\partial t} + \frac{1}{2} v S^2 \frac{\partial^2 V}{\partial S^2} + \rho v \beta S \frac{\partial^2 V}{\partial v \partial S} + \frac{1}{2} v \beta^2 \frac{\partial^2 V}{\partial v^2} + rS \frac{\partial V}{\partial S} - rV& \\
= -(\alpha - \phi\beta\sqrt{v})\frac{\partial V}{\partial v},& \qquad (6)\\
V(S, v, T) = \varphi(S, v),& \qquad (7)
\end{aligned}
$$

where $\phi = \phi(S, v, t)$ is a function of the independent variables S, v and t, which does not depend on the specific claim.

main derivation steps

First recall multivariate Ito's formula (Hull, appendix of Chapter "Wiener Process and Ito's Lemma"):

Let $\boldsymbol{X} = (X_1, ..., X_n)^{\mathsf{T}}$ be a multidimensional Ito process, i.e.

$$d\boldsymbol{X} = \boldsymbol{\mu} dt + \boldsymbol{\sigma} d\boldsymbol{W},$$

where

- $\boldsymbol{W}$ is a d-dimensional Wiener process with correlation matrix $\boldsymbol{\rho}$
- $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are vector and matrix processes.

Let $f : [0, \infty) \times \mathbb{R}^n \to \mathbb{R}$ be a $C^{1,2}$ function. Then

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dX_i + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j} dX_i dX_j.$$

with the "operational rules":

$$dtdt = 0, \qquad dtd\boldsymbol{W} = d\boldsymbol{W}dt = 0, \qquad dW_i dW_j = \rho_{i,j} dt \tag{8}$$

Start mimicking the risk-less portfolio strategy of LV models

- Setup a portfolio $\Pi = V - \Delta S - \Delta^1 V^1$ where $V^1(S_t, v_t, t)$ is the price process of an alternative derivative instrument.
- Compute the dynamics

$$d\Pi = dV - \Delta dS - \Delta^1 dV^1 \tag{9}$$

apply Ito's formula to V and V^1 to obtain

$$\begin{aligned} dV &= V_t dt + V_s dS + \frac{1}{2} vS^2 V_{ss} dt + V_v dv + \frac{1}{2} v\beta^2 V_{vv} dt + \rho vS\beta V_{vS} dt \\ dV^1 &= V^1_t dt + V^1_s dS + \frac{1}{2} vS^2 V^1_{ss} dt + V^1_v dv + \frac{1}{2} v\beta^2 V^1_{vv} dt + \rho vS\beta V^1_{vS} dt \end{aligned}$$

- Inserting dV and dV^1 into (9) we see that to eliminate risk from $d\Pi$ we need to set

$$\begin{aligned} V_s - \Delta - \Delta^1 V^1_s &= 0 \qquad \text{to eliminate risk coming from } dS \\ V_v - \Delta^1 V^1_v &= 0 \qquad \text{to eliminate risk coming from } dv \end{aligned}$$

With $\Delta^1 = V_v / V_v^1$ and $\Delta = V_s - \Delta^1 V_s^1$ the dynamics $d\Pi$ become

$$d\Pi =$$

$$\left(V_t + \frac{1}{2} v S^2 V_{ss} + \frac{1}{2} v \beta^2 V_{vv} + \rho v S \beta V_{vS} - \Delta^1 \left(V_t^1 + \frac{1}{2} v S^2 V_{ss}^1 + \frac{1}{2} v \beta^2 V_{vv}^1 + \rho v S \beta V_{vS}^1 \right) \right) dt$$

$$= r \left(V - (V_s - \Delta^1 V_s^1) S - \Delta^1 V^1 \right) dt \, ,$$

where the second equality must hold in virtue of the no arbitrage assumption.

Substituting the expression for Δ^1 and collecting terms in V and V^1 we obtain that the following must hold

$$\frac{V_t + \frac{1}{2} v S^2 V_{ss} + \frac{1}{2} v \beta^2 V_{vv} + \rho v S \beta V_{vS} + rSV_s - rV}{V_v} = \frac{V_t^1 + \frac{1}{2} v S^2 V_{ss}^1 + \frac{1}{2} v \beta^2 V_{vv}^1 + \rho v S \beta V_{vS}^1 + rSV_s^1 - rV^1}{V_v^1}$$

The only way the equality above can hold for arbitrary V and V^1 is if there exists a function $f(S, v, t)$ which only depends on the independent variables S,v, and t such that for any pricing function V it holds that

$$\frac{V_t + \frac{1}{2} v S^2 V_{ss} + \frac{1}{2} v \beta^2 V_{vv} + \rho v S \beta V_{vS} + rSV_s - rV}{V_v} = f(S, v, t) \, .$$

The Market price of volatility risk-1

- The function f is not determined within the SV model, but must be specified *exogenously* just as we have to specify the coefficients α and β in the variance dynamics.

- So - in contrast to LV models - the price of derivatives in SV models is not uniquely determined by the model specification (10)-(11), but there exists infinitely many price processes consistent with the no-arbitrage assumption.

- In particular, a market described by SV dynamics is *incomplete*: derivative assets are not necessarily hedge-able via replicating portfolios based on the underlying asset S and the bank account B.

- Without loss of generality, we can re-write the function f as follows

$$f = -(\alpha - \beta\sqrt{v}\phi)$$

  where the process $\lambda_t = \phi(S_t, v_t, t)$ is known as the *the market price of volatility risk*- as it represents the expected excess return due to the presence of volatility risk

The Market price of volatility risk-2

Elaboration on black-board (pp 6-7)

Risk-neutral valuation

An application of the multivariate Feynman-Kac theorem yields that the pricing function solving (6) - (7) can be represented as follows

$$V(s, v, t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\varphi(S_T, v_T)|S_t = s, v_t = v],$$

where the $\mathbb{Q}$-dynamics of (S, v) are given by

$$dS = rSdt + \sqrt{v}SdZ_1^{\mathbb{Q}} \quad (10)$$

$$dv = (\alpha - \phi\beta\sqrt{v})dt + \beta\sqrt{v}dZ_2^{\mathbb{Q}} \quad (11)$$

where $dZ_1 dZ_2 = \rho dt$ and we have omitted all the dependence on (t, v, S) on the coefficients α, β and the market price of the volatility risk ϕ.

We see that as long as we are concerned with derivatives pricing, it is irrelevant ho exactly we specify α and ϕ *per se*.

The object - apart from β -that really determines the price of derivatives is the risk-neutral drift $\alpha - \phi\beta$. In other words- for pricing derivatives - all we need to do is to specify the risk-neutral dynamics of the variance process v.

And that's exactly what we will do from now on!

The Local Vol of a Stochastic Vol model

- Before moving on to examine the details of some well known SV models, let's establish the connection between the volatility - the variance- process and the local volatility associated to a SV model.

- The result is as follows:

  Proposition Let $r = d = 0$ and let $\sigma^2(K, T, S_0)$ be the squared local volatility associated to the risk-neutral model

$$dS = \sqrt{v} S dZ_1^{\mathbb{Q}} \tag{12}$$
$$dv = \alpha dt + \beta \sqrt{v} dZ_2^{\mathbb{Q}} . \tag{13}$$

  Then it holds that

$$\sigma^2(K, T, S_0) = \mathbb{E}^{\mathbb{Q}}[v_T | S_T = K] \tag{14}$$

  proof On blackboard (pp13-14).

Classic SV models

Classic SV models

- A quite popular subclass of **RISK-NEUTRAL** SV model takes the following form

$$dS_t = (r-q)S_t dt + \sqrt{v_t} S_t dB_t \tag{15}$$

$$dv_t = a(v_t)dt + b(v_t)dW_t \tag{16}$$

  with correlation $dB_t dW_t = \rho dt$.

- Here we also have assumed that the underlying asset pays dividends at constant yield q. At some point we will set, without loss of generality, $r = q = 0$.

- We shall focus on the time homogeneous case, but the time inhomogeneous specification

$$dv_t = a(v_t, t)dt + b(v_t, t)dW_t$$

  does not present particular complications.

Well known examples

- Hull and White (1987):

$$dv_t = \mu v_t dt + \xi v_t dW_t$$

- Heston (1993)

$$dv_t = \lambda(\theta - v_t)dt + \varepsilon\sqrt{v_t}dW_t$$

- Vasicek?

$$dv_t = \lambda(\theta - v_t)dt + \varepsilon dW_t$$

  NOT FEASIBLE!

- But

$$v_t = v_0 \exp X_t \qquad \text{where } dX_t = \lambda(\theta - X_t)dt + \varepsilon dW_t$$

  is perfectly legal. It is (somewhat) known as the Bergomi model

- 3/2 model - Heston (1997) and Lewis (2000)

$$dv_t = \lambda v_t(\theta - v_t)dt + \varepsilon(v_t)^{3/2}dW_t$$

- Continuous Garch model- Lewis (2000)

$$dv_t = \lambda(\theta - v_t)dt + \varepsilon v_t dW_t$$

Back to the general case

$$\begin{aligned} dS_t &= (r-q)S_t dt + \sqrt{v_t} S_t dB_t \\ dv_t &= a(v_t)dt + b(v_t)dW_t \end{aligned}$$

- As the model is specified directly under a risk-neutral measure, the corresponding arbitrage free price $C(S_0, v_0, K, T)$ of a call option striking at K and maturing at T is given by

$$C(S_0, v_0, K, T) = e^{-rT} E[(S_T - K)^+].$$

- Main idea: for each path of v, the model is a like BS. Then we may try to express the SV price as an average of the BS formula over all the possible paths of v

First reformulate in terms of independent BMs

- We start by writing the solution for the price process at T

$$S_T = e^{(r-q)T} S_0 e^{-\frac{1}{2}\int_0^T v_s ds + \int_0^T \sqrt{v_s} dB_s} \tag{17}$$

- Expressing $B_t = \rho W_t + \sqrt{1-\rho^2} Z_t$ with $dW_t dZ_t = 0$, plugging in (17) and re-arranging we get

$$S_T = e^{(r-q)T} S_0 \, e^{-\frac{1}{2}\int_0^T \rho^2 v_s ds + \int_0^T \rho\sqrt{v_s} dW_s} \, e^{-\frac{1}{2}\int_0^T (1-\rho^2) v_s ds + \int_0^T \sqrt{(1-\rho^2) v_s} dZ_s}$$

$$= e^{(r-q)T} S_t^{eff} e^{-\frac{1}{2}\int_0^T (1-\rho^2) v_s ds + \int_0^T \sqrt{(1-\rho^2) v_s} dZ_s} \tag{18}$$

where we have introduced the "effective spot" S_t^{eff}

$$S_T^{eff} = S_0 \exp\left(-\frac{1}{2}\int_0^T \rho^2 v_s ds + \int_0^T \rho\sqrt{v_s} dW_s\right) \tag{19}$$

- Notice that we have separated the quantities depending of the two independent BM Z and W.

(Generalized) Mean-variance Mixture Returns

- Now introduce also the "effective variance" v_t^{eff} defined as

$$v_T^{eff} = \frac{(1-\rho^2)}{T} \int_0^T v_s ds \tag{20}$$

- Then the distribution of the log-spot $X_T = \log S_T$ at maturity is given by a (generalized) mean-variance mixture of a normal distribution - similarly to the variance gamma distribution-

$$X_T \sim (r-q)T + \log S_T^{eff} - \frac{1}{2} T v_T^{eff} + \sqrt{T v_T^{eff}} N(0,1)$$

with S_T^{eff}, v_T^{eff} independent of $N(0,1)$

- Notice the similarity with BS where

$$X_T \sim (r-q)T + \log S_0 - \frac{1}{2} T\sigma^2 + \sqrt{T\sigma^2}\, N(0,1)$$

is simply a normal distribution

SV models vs BS model

- **Theorem** (Romano and Touzi, 1997): Let $C(S_0, v_0, K, T)$ be the call price under the SV model (15) and (16). Let $C_{BS}(S_0, \sigma, K, T)$ denote the BS price of the same option corresponding to a volatility value σ. Then

$$C(S_0, v_0, K, T) = E[C_{BS}(S_T^{eff}, \sqrt{v_T^{eff}}, K, T)].$$

- In words: the option value under SV is a weighted sum or a mixture of the BS values with an effective spot and effective variance. The effective values depend **only** upon the variance process. So the pricing problem is reduced to take expectations over the variance process alone.

- Notice that S_t^{eff} is a (local) martingale

Pricing via simulation

- Summing up:

$$C(S_0, v_0, K, T) = E[C_{BS}(S_T^{eff}, \sqrt{v_T^{eff}}, K, T)]$$

  with

$$S_T^{eff} = S_0 \exp\left(-\frac{\rho^2}{2}\int_0^T v_s ds + \rho\int_0^T \sqrt{v_s} dW_s\right)$$

$$v_T^{eff} = \frac{(1-\rho^2)}{T}\int_0^T v_s ds$$

  and v_t evolving as

$$dv_t = a(v_t)dt + b(v_t)dW_t$$

- How can we evaluate the expectation above? Unless we know the exact joint distribution of (S_T^{eff}, v_T^{eff}), the first attempt is to proceed via simulation.
- Notice that we need to simulate only the variance paths.

Simulating (S_T^{eff}, v_T^{eff}) via Euler discretization

- In fact we need to simulate the two following integrals:

$$I_T^1 = \int_0^T v_s ds \qquad I_T^2 = \int_0^T \sqrt{v_s} dW_s$$

- The dynamics are given by

$$dI_t^1 = v_t dt, \quad I_0^1 = 0 \quad \text{and} \quad dI_t^2 = \sqrt{v_t} dW_t, \quad I_0^2 = 0$$

- So discretizing the time interval in N points and mesh $\Delta t = T/N$ we get, for $i = 0, \ldots, N-1$

$$I_{i+1}^1 = I_i^1 + v_i \Delta t \qquad I_{i+1}^2 = I_i^2 + \sqrt{\Delta t v_i} \Phi$$

  with $\Phi \sim N(0,1)$ sampled from a Standard Normal Variable.

- As for the variance process, we could introduce (at first) a simple Euler scheme

$$v_{i+1} = v_i + a(v_i)\Delta t + b(v_i)\sqrt{\Delta t}\Phi$$

The (primitive) Algorithm

The sketched algorithm to simulate one draw from (S_T^{eff}, v_T^{eff}) is as follows

1. Initialize current $I^1 = 0$, $I^2 = 0$ and $v = v_0$ as where v_0 is the initial variance value.
2. for $i = 1 : N$
   - generate Φ from $N(0,1)$;
   - update

$$I^1 = I^1 + v\Delta t; \quad I^2 = I^2 + \sqrt{v\Delta t}\Phi; \quad v = v + a(v)\Delta t + b(v)\sqrt{\Delta t}\Phi$$

   end

3. Set $S^{eff} = S_0 \exp\left(-\frac{1}{2}\rho^2 I^1 + \rho I^2\right)$; and $v^{eff} = \frac{1}{T}(1-\rho^2)I^1$.

Finally, the option price is obtained by taking the arithmetic average

$$C(S_0, v_0, T) = \sum_{j=1}^{Nsim} [C_{BS}(S_j^{eff}, \sqrt{v_j^{eff}}, T)]/Nsim$$

over a large number $Nsim$.

Zero correlation

- If there is no correlation, i.e. $\rho = 0$, then the effective spot coincides with the actual spot price

$$S_T^{eff} = S_0 e^{-\frac{1}{2}\int_0^T \rho^2 v_s ds + \int_0^T \rho\sqrt{v_s} dW_s} = S_0$$

- In this case the variance process v determines the option price uniquely through the distribution of the integrated variance (annualized)

$$v_T^{eff} = \frac{1}{T}\int_0^T v_s ds$$

- This simpler case was studied in Hull and White (1987), one of the first papers on SV.

- In what follows, we focus on the $\rho = 0$ case, and we use the mixing representation to get some insights on the implied volatility surface associated with a SV model.
- For simplicity, we also assume an economy with zero interest rates and dividend yields
$$r = q = 0.$$
The extension to constant r and q is left as an exercise
- Also, with a slight abuse of notation, we express the BS formula in terms of TOTAL variance $V = \sigma^2 T$ and log-moneyness $x = \log K/S$
$$C_{BS}(S, V, T, K) = C_{BS}(S, V) = S\,N(d_1) - K\,N(d_2),$$
where
$$d_{1,2} = -\frac{x}{\sqrt{V}} \pm \frac{\sqrt{V}}{2}. \tag{21}$$

A simple Approximation (Hull&White /Drimus/Nicolato&Sloth)

- Let's start by fixing a maturity T. Since $\rho = 0$, the mixing rep for the true SV call price becomes
$$C(S_0, T) = E[(S_T - K)^+] = E[C_{BS}(S_0, V_T)], \tag{22}$$
where $V_T = \int_0^T v_s ds$.
- Let's do the simplest possible thing: apply a formal Taylor expansion of C_{BS} around $\mathbb{E}V_T$ and then take expectations

Few observations

$$C(S_0, T) \approx C_{BS}(S_0, \mathbb{E}V_T) + \frac{1}{2}\mathbb{E}[(V_T - \mathbb{E}V_T)^2]\frac{\partial^2 C_{BS}}{\partial v^2}(S_0, \mathbb{E}V_T) + \frac{1}{3!}\dots \quad (23)$$

- The approximation given in (23) is meaningful only under the assumption that all the moments $\mathbb{E}[(V_T - \mathbb{E}V_T)^n]$ entering the expression are finite.
- Also, the practical applicability relies on possibility of computing $\mathbb{E}[V_T^n]$, preferably in closed form.
- Finally, notice that even when all the moments are finite, there is no guarantee that the expansion (23) converges with higher-order approximations capturing the true option prices more accurately. A Taylor expansion of the BS price in the variance has a finite radius of convergence, while here we expand the mixing representation over the entire domain $V \in (0, \infty)$.
- Still, let's truncate the approx at the first meaningful order $n = 2$ and let's see which sort of information it conveys

Second order approx

$$C(S_0, T) \approx C_{BS}(S_0, \mathbb{E}V_T) + \frac{1}{2} Var[V_T] \frac{\partial^2 C_{BS}}{\partial V^2}(S_0, \mathbb{E}V_T) \tag{24}$$

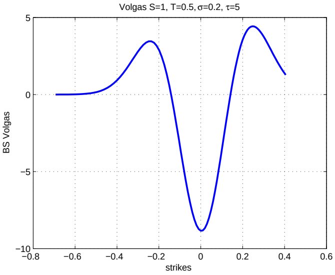
where

$$\begin{aligned} Volga = \frac{\partial^2 C_{BS}}{\partial V^2}(S, V) &= \frac{S}{4V^{3/2}} N'(d_1)(d_1 d_2 - 1) \\ &= \frac{S}{4V^{3/2}} N'(d_1) \left(\frac{x^2}{V} - \frac{V}{4} - 1 \right) \end{aligned} \tag{25}$$

What can be deduced from this expression?

Approx - Illustration -1

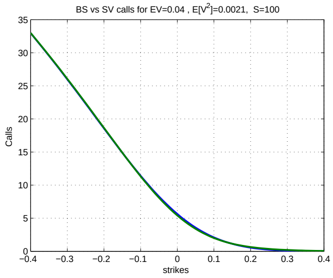
For a fixed value V of variance, $\frac{\partial^2 C_{BS}}{\partial V^2}$ plotted against $x = \log K/S$ looks like

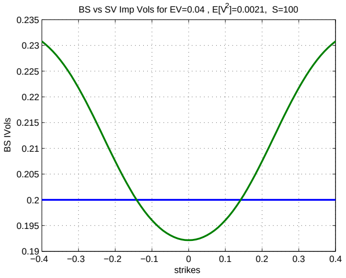


- Around ATM $\frac{\partial^2 C_{BS}}{\partial V^2} < 0$, while for deep ITM/OTM $\frac{\partial^2 C_{BS}}{\partial V^2} > 0$.
- We see from (25) that this is independent of the particular value V chosen - although the points of shift will change. So it holds also for $V = \mathbb{E}[V_T]$.
- Hence, the approx (24) captures the fact that for any distribution of V_T, BS model underprices options for "deep" ITM/OTM and overprices around ATM

Approx - Illustration -2

Let's now compare the SV approximation for arbitrary $\mathbb{E}V_T$, $\mathbb{E}[V_T^2]$ with the corresponding BS prices





- The approx captures the smile, which furthermore seems to be a symmetric function of log-monyness $x = \log K/S$ centered around the ATM point $x = 0$.
- This is in fact a characteristic of the smile associated with a SV with zero correlation $\rho = 0$. Let's formalize.

Smile symmetry for no leverage $\rho = 0$

Theorem (Renault and Touzi (1996))

Consider the risk neutral SV model (15), (16)*, with* $\rho = 0$*. Let* $\sigma_{BS}(x, v)$ *denote the implied volatility corresponding to* $x = \log K/S$ *and current level of (instantaneous) variance* v*. Then*

$$\sigma_{BS}(x, v) = \sigma_{BS}(-x, v)$$

Before moving on to the proof, let's point out that the implied vol $\sigma_{BS}(x, v)$ (at a fixed maturity) does in fact depend on current spot S and K only trough the ratio K/S

In fact the true call price can be expressed as follows

$$C(S, v, K, T) = S\mathbb{E}[(S_T/S - e^x)^+]$$

where the distribution of S_T/S depends on current value v of variance process but does not depend on S.

- Hence, we can define the implied vol σ_{BS} as the unique solution of

$$\frac{C(S, v, K, T)}{S} = \frac{C_{BS}(S, \sigma_{BS}^2 T, K, T)}{S}$$

$$\Rightarrow \frac{C(S, v, K, T)}{S} = N(d_1(x, \sigma_{BS}^2 T)) - e^x N(d_2(x, \sigma_{BS}^2 T)) \tag{26}$$

where

$$d_{1/2}(x, V) = -\frac{x}{\sqrt{V}} \pm \frac{\sqrt{V}}{2} \tag{27}$$

- Since the LHT in (26) depends only on (x, v), and RHT depends only on x, we see that the implied volatility σ_{BS}^2 depends uniquely on (x, v).

Proof -1

- Let H denote the normalized Black-Scholes function

$$H(x,V) = N(d_1(x,V)) - e^x N(d_2(x,V)) \qquad \text{for } (x,V) \in \mathbb{R} \times \mathbb{R}_+$$

  where $d_{1/2}$ are defined as in (27).

- Few calculations show that for any (x,V) it holds that

$$H(-x,V) = (1 - e^{-x}) + e^{-x} H(x,V) \tag{28}$$

- Notice that, in virtue of the mixing representation (22), the true normalized call price is given by

$$\frac{C(S,v,K,T)}{S} = \mathbb{E}[H(x,V_T)] \qquad where\, V_T = \int_0^T v_s ds.$$

1. So, by using the property (28) we get

$$\mathbb{E}[H(-x, V_T)] = (1 - e^{-x}) + e^{-x}\mathbb{E}[H(x, V_T)] \tag{29}$$

2. On the other hand

$$\begin{aligned} \mathbb{E}[H(-x, V_T)] &= H(-x, T\sigma^2_{BS}(-x, v)) \\ &= (1 - e^{-x}) + e^{-x}H(x, T\sigma^2_{BS}(-x, v)) \end{aligned} \tag{30}$$

3. Equating (29) and (30) $\quad \Rightarrow \quad \mathbb{E}[H(x, V_T)] = H(x, T\sigma^2_{BS}(-x, v))$

   but by definition of imp vol $\quad \Rightarrow \quad \mathbb{E}[H(x, V_T)] = H(x, T\sigma^2_{BS}(x, v))$

4. By uniqueness of imp vol $\quad \Rightarrow \quad \sigma^2_{BS}(x, v) = \sigma^2_{BS}(-x, v)$ $\square$

Back to our simple approx

- Let's get back to the simple 2nd order approx. It seems to display some nice properties, e.g. it does give us a smile, and it captures symmetry (you could try to check it out actually).
- But the main question is: does it work when implemented for specific dynamics?
- Hull & White check out the log normal dynamics $dv = \mu v dt + \xi v dW$ and provide expressions for the moments of integrated variance.
- How about the Heston model? To implement the 2nd order approx, we need to be able to compute $\mathbb{E}[V_T]$ and $\mathbb{E}[V_T^2]$ for $V_T = \int_0^T v_s ds$ when v follows the CIR ´dynamics

$$dv = \kappa(\theta - v)dt + \varepsilon\sqrt{(}v)dt \tag{31}$$

From Dufresne (2001), "The Integrated Square-Root Process"

For the square root process

$$dX = (\alpha X + \beta)\, dt + \gamma\sqrt{X} dW, \qquad X_0 = \bar{x} > 0$$

the following holds

- Let $m_k(t) = \mathbb{E}X_t^k$. Then $m_k(t)$, $k = 0, 1, \ldots$ solve the ODEs

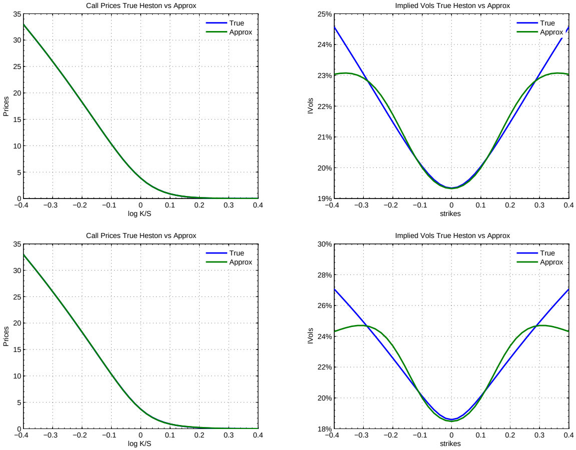
$$m_0(t) = 1 \qquad t \geq 0$$
$$m_k'(t) = a_k m_k(t) + b_k m_{k-1}(t), \qquad m_k(0) = \bar{x}^k$$

  with $a_k = \alpha k$, $b_k = \beta k + \frac{1}{2}\gamma^2 k(k-1)$.

- Let $M_{jk}(t) = \mathbb{E}Y_t^j X_t^k$, where $Y_t = \int_0^t X_s ds$. Then $M_{jk}(t)$ solve the ODEs

$$M_{jk}'(t) = a_k M_{jk}(t) + b_k M_{j,k-1}(t) + jM_{j-1,k+1}$$

Some Illustrations/Results



Top panel: $\kappa = 1.15$, $\theta = 0.04$, $v_0 = \theta$, $\epsilon = 0.4$ $\rho = 0$, and $T = 0.25$
Bottom panel: $\kappa = 1.15$, $\theta = 0.04$, $v_0 = \theta$, $\epsilon = 0.6$ $\rho = 0$, and $T = 0.25$

The fundamental transform

Laplace transform of log-price

For simplicity, assume $r = d = 0$ and consider a risk-neutral SV model of the type (15)-(16). Reformulated in terms of log-price $X_t = \log S_t$, the dynamics become

$$dX = -\frac{1}{2}vdt + \sqrt{v}dB \tag{32}$$

$$dv = a(v)dt + b(v)dW, \tag{33}$$

where $dBdW = \rho dt$.

In what follows we focus on the Laplace transform of the log-price $X_T = \log S_T$

$$\mathcal{L}(u; T) = \mathbb{E}[e^{uX_T}] \tag{34}$$

defined in the domain

$$\mathcal{D} = \{u \in \mathbb{C} : \mathbb{E}[|e^{uX_T}|] < \infty\}.$$

- The transform $\mathcal{L}$ is a crucial quantity for a number of reasons.
- Setting $u = iz$ yields the characteristic function (cf)of the log-price
$$\mathcal{L}(iz, T) = \mathbb{E}[e^{izX_T}].$$
- Characteristic functions are in one-to-one correspondence with densities. For a number of models the cf of the risk-neutral log-price is explicitly computable - or more tractable than the density itself which instead remains a rather difficult object to deal with.
- Cfs of log-prices are at the core of Fourier pricing, an efficient and accurate numerical method that you will study with Thomas.
- As as side note, under enough regularity we may use $\mathcal{L}$ to compute moments as
$$\mathbb{E}[X_T^n] = \frac{\partial \mathcal{L}^n}{\partial u^n}(0; T)$$
.

The PDE for the Laplace transform of the log-price

- Our task now is to compute $\mathcal{L}$.
- In fact, rather than considering the "unconditional" expectation, we shall consider the conditional Laplace transform of the log-price X_T. That is - with a slight abuse of notation - we will compute

$$\mathcal{L}(u; t, T) = \mathbb{E}\left[e^{uX_T} | \mathcal{F}_t\right]$$

  Clearly the unconditional Laplace transform is obtained by setting $t = 0$.
- Fix T and u and consider the process $M_t = \mathcal{L}(t, T; u)$. Then
  - By construction M_t is a martingale such that $M_T = e^{uX_T}$
  - Because of the markovianity of (X, v), M_t is a transformation of v_t and X_t, i.e. - abusing once again the notation-

$$\mathbb{E}\left[e^{uX_T} | \mathcal{F}_t\right] = \mathbb{E}\left[e^{uX_T} | X_t, v_t\right] = \mathcal{L}(X_t, v_t, t, T; u)$$

- Apply Ito's formula to compute the dynamics of $M_t = \mathcal{L}(X_t, v_t, t, T; u)$. We obtain

$$dM = \mathcal{L}_t dt + \mathcal{L}_X dX + \mathcal{L}_v dv + \frac{1}{2}\mathcal{L}_{XX}(dX)^2 + \frac{1}{2}\mathcal{L}_{vv}(dv)^2 + \mathcal{L}_{Xv}(dXdv)$$

$$= \left(\mathcal{L}_t - \frac{1}{2}v\mathcal{L}_X + a(v)\mathcal{L}_v + \frac{1}{2}v\mathcal{L}_{XX} + \frac{1}{2}\mathcal{L}_{vv}b^2(v) + \mathcal{L}_{Xv}\sqrt{v}b(v)\rho\right) dt + \tag{35}$$

$$+ \mathcal{L}_X\sqrt{v}dB + \mathcal{L}_v b(v)dW$$

- Since M is a martingale, the drift in (35) must be equal to 0, yielding the following PDE

$$\mathcal{L}_t - \frac{1}{2}v\mathcal{L}_X + a(v)\mathcal{L}_v + \frac{1}{2}v\mathcal{L}_{XX} + \frac{1}{2}b^2(v)\mathcal{L}_{vv} + \sqrt{v}b(v)\rho\mathcal{L}_{Xv} = 0 \tag{36}$$

with final condition $\mathcal{L}(X, v, T, T; u) = e^{uX}$.

The PDE for the Laplace transform of the log-price

- Also notice that $\mathcal{L}$ takes the form

$$\mathcal{L}(X_t, v_t, t, T; u) = \mathbb{E}\left[e^{uX_T}|\mathcal{F}_t\right] = e^{uX_t}\mathbb{E}\left[e^{u(X_T - X_t)}|\mathcal{F}_t\right] \tag{37}$$

where – under the SV dynamics (32)-(33) – the quantity $e^{u(X_T - X_t)}$ does not depend in any way from X_t.

- Let's introduce some notation for the normalized conditional Laplace transform of the log-price

$$H(v, t, T; u) = \mathbb{E}\left[e^{u(X_T - X_t)}|v_t = v\right] \tag{38}$$

- H is known as the *fundamental transform* for the model dynamics (32)-(33).

- So $\mathcal{L}$ must solve

$$\mathcal{L}_t - \frac{1}{2} v \mathcal{L}_X + a(v)\mathcal{L}_v + \frac{1}{2} v \mathcal{L}_{XX} + \frac{1}{2} b^2(v)\mathcal{L}_{vv} + \sqrt{v}\, b(v)\rho \mathcal{L}_{Xv} = 0$$
$$\mathcal{L}(X, v, t, T; u) = e^{uX}$$

- But $\mathcal{L} = e^{uX} H(v, t, T; u)$ and therefore

$$\mathcal{L}_t = e^{uX} H_t, \qquad \mathcal{L}_X = u e^{uX} H, \qquad \mathcal{L}_{XX} = u^2 e^{uX} H,$$

$$\mathcal{L}_v = e^{uX} H_v, \qquad \mathcal{L}_{vv} = e^{uX} H_{vv}, \qquad \mathcal{L}_{Xv} = u e^{uX} H_v$$

- Inserting the derivatives in the PDE we obtain that the normalized transform H must satisfy the boundary value problem

$$H_t + \frac{u^2 - u}{2}\, vH + \left(a(v) + u\rho\sqrt{v}\, b(v)\right) H_v + \frac{1}{2} b^2(v) H_{vv} = 0 \qquad (39)$$
$$H(v, T, T; u) = 1. \qquad (40)$$

- Notice that the bvp (39)-(40) depends only on the variables v and t. Once again we notice a dimensionality reduction wrt the more general class of SV models (see e.g. (6)-(7)).

The fundamental transform in the Heston model

Consider the risk-neutral Heston model

$$dX = -\frac{1}{2}vdt + \sqrt{v}dB$$
$$dv = \lambda(\theta - v)dt + \varepsilon\sqrt{v}dW,$$

where $dBdW = \rho dt$. The fundamental transform $H(v, t, T; u) = \mathbb{E}\left[e^{u(X_T - X_t)}|v_t = v\right]$ takes the following form

$$\mathbb{E}\left[e^{u(X_T - X_t)}|v_t = v\right] = e^{A(T-t,u)+vB(T-t,u)}, \tag{41}$$

where the functions $A(\tau, u)$, $B(\tau, u)$ satisfy the Riccati ODEs

$$\frac{\partial A}{\partial \tau} = \lambda\theta B \tag{42}$$
$$\frac{\partial B}{\partial \tau} = -(\lambda - u\varepsilon\rho)B + \frac{1}{2}\varepsilon^2 B^2 + \frac{u^2 - u}{2} \tag{43}$$

with initial conditions $A(0, u) = B(0, u) = 0$.

Blackboard

- In fact the solution to (42) and (43) can be written down in closed form.
- For u such that $Re\left((\lambda - u\rho\varepsilon)^2 - \varepsilon^2(u^2 - u)\right) > 0$ they read as follows

$$\begin{aligned} A(\tau, u) &= \frac{2\lambda\theta}{\varepsilon^2} \log\left(\frac{\gamma e^{(\lambda - u\rho\varepsilon)\,\tau/2}}{\sinh(\gamma\,\tau/2)\left(\lambda - u\rho\varepsilon + \gamma \coth(\gamma\,\tau/2)\right)}\right) \\ B(\tau, u) &= \frac{u^2 - u}{\lambda - u\rho\varepsilon + \gamma \coth(\gamma\,\tau/2)} \end{aligned}$$

where

$$\gamma = \sqrt{(\lambda - u\rho\varepsilon)^2 - \varepsilon^2(u^2 - u)}\,.$$

- To obtain the cf set $u = iz$ with $z \in \mathbb{R}$.

The Volatility of Volatility Expansion

- For a selection of model, the transform H is available in closed form and the price $C(S, v, \tau, K)$ of a call with time-to-maturity τ, strike K when the current spot price is S and the current variance value is v can be computed via a Fourier method of choice (Carr-Madan, Lewis, Lipton), e.g.,

$$C(S, v, \tau, K) = S - \frac{K}{2\pi} \int_{iz_i-\infty}^{iz_i+\infty} e^{-iz\kappa} H(\tau, v; -iz) \frac{1}{z^2 - iz} dz_r \qquad (44)$$

  where $\kappa = \log(S/K)$. (Recall we are in the simplified economy with zero interest rates and dividends yields).

- The implied volatility can then be computed as usual by inversion of Black-Scholes formula.

The Volatility of Volatility Expansion

- However, there are two problems/issues with the formula above:
  - What if H is not available?
  - The formula works very much as a black-box and does not provide any transparent insights on the effects of the specific dynamics of the variance process v on the implied volatility surface.
- There is in fact a vast literature aiming to find a tractable approximation for the fundamental transform H which:
  - Allows for efficient pricing when H is not available.
  - Allows for neat interpretations.
  - Allows for explicit implied volatility approximations which bypass the numerical inversion.
- Not at all an easy task!

The short time-to-expiry limit

- It becomes simpler if one focusses on certain regions of the vol surface- typically asymptotic results at short or long maturities
- Lots of the analysis focusses in the short-mauturity/small moneyness - partly in virtue of data availability
- By approximating the solution of the PDE (39)- it can be shown that near at-the-money (ATM) - the short-time-to-expiry limit $\sigma^2_{BS}(x, 0, v)$ -here parametrized in log moneyness $x = \log(K/S)$ is approx given by

$$\sigma^2_{BS}(x, 0, v) \approx v + \frac{1}{2}\frac{\rho b(v)}{\sqrt{v}}x + \left[\left(\frac{1}{12} - \frac{11}{48}\rho^2\right)\frac{b^2(v)}{v^2} + \frac{\rho^2}{6v}b(v)\frac{\partial b}{\partial v}\right]x^2 + O(x^3) \tag{45}$$

(for small moneyness $|x|$ values.)

The short time-to-expiry limit - 2

What can we learn from expression (45)?

- We see that the approx ATM implied variance converges to the instantaneous spot variance v, which can be shown to be **the limit of the exact quantity**
- Also, define the ATM implied volatility skew as

$$Sk(\tau) = \frac{\partial \sigma_{BS}(x, \tau, v)}{\partial x}\Big|_{x=0} \tag{46}$$

- From (45) we see that the short-time-to-expiry limit of the approx skew $Sk(0)$ takes the following form

$$Sk(0) \approx \frac{1}{4}\frac{\rho b(v)}{v} \tag{47}$$

  which once again correspond to the exact quantity.
  A couple of comments on this: **Blackboard**
- The same holds for the ATM short-time-to-expiry limit of the approx convexity.

Heston: Limitations

- It seems that the Heston model is not quite capable to capture the behavior of the Skew $\frac{\partial \sigma_{BS}^2(x,\tau)}{\partial x}$ for short maturities.
- For short maturities, the empirically observed skew seems to be much steeper than the skew generated by Heston model (or any other SV model, actually)

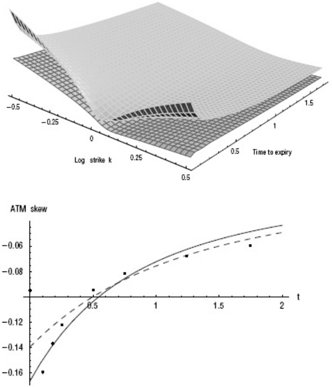


FIGURE 3.4 Graph of SPX ATM skew versus time to expiry. The solid line is a fit of the approximate skew formula (3.21) to all empirical skew points except the first; the dashed fit excludes the first three data points.

- Hence the need of introducing new models beyond SV